Reg. No. | | | | | | | | | | |

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER B.TECH. (INSTRUMENTATION AND CONTROL ENGG.)

END SEMESTER DEGREE EXAMINATIONS, DECEMBER - 2019

SUBJECT: DIGITAL IMAGE PROCESSING [ICE 4021]

TIME: 3 HOURS MAX. MARKS: 50

Instructions to candidates :*Answer ALL questions and missing data may be suitably assumed.*

1A. Illustrate the structure of image formation model. Give the significance of various levels in image processing.

1B. Discuss the sampling and quantization process with an example each.

1C. Write a short note on optical illusions.

(4+4+2)

2A. Describe the followings with an example,

i) Connectivity of pixels ii) Euclidean distance

2B. Apply the 3×3 average mask using zero padding to an image shown below.

1	2	4	5
5	2	5	2
1	1	3	6
2	4	6	7

2C. Explain and compare histogram equalization and matching using suitable example.

(3+3+4)

3A. Discuss any two order statistics filter with its advantage.

3B. Explain the following with relevant graphs,

i) Gray level slicing ii) Gamma correction

3C. With a neat block diagram, explain the steps involved in filtering approach that uses illumination-reflectance model in its operation.

(2+3+5)

4A. Explain any one method that addresses interpixel redundancy with an example.

4B. Describe loss-less predictive coding model with its processing stages.

4C. Encode the sequence $a_1\ a_2\ a_3\ a_3\ a_4$ using following information.

Source Symbol	Probability
a_1	0.2
a_2	0.2
a_3	0.4
a_4	0.2

(2+4+4)

5A. With mathematical formulation explain bottom-hat transformation.

5B. Explain segmentation scheme based on the colour and shape of an object.

5C. With a neat block diagram, explain the stages required to recognize the face automatically.

(2+4+4)

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